

Learning ML, DL, NLP
course

①

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Sec - 27

Concepts of Linear Regression

Lec-1 Simple Linear
Regression Introduction

Lec-2 Understanding
Equations

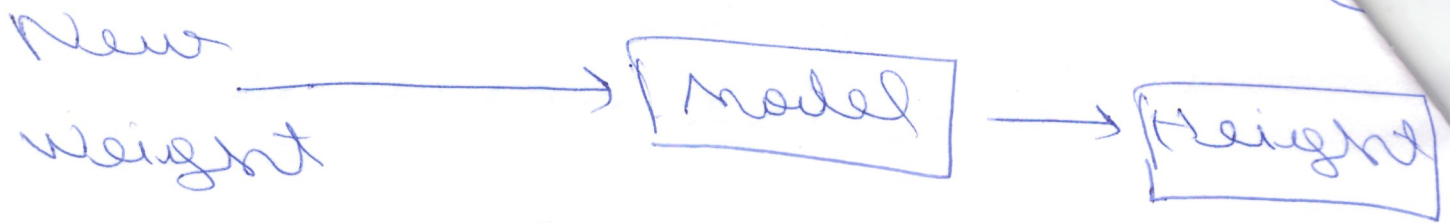
Lec-3 Cost Function

Simple Linear Regression

Supervised ML $\rightarrow$ Regression

Dataset :- $\rightarrow$ I/P Features $\rightarrow$ O/P Features

Weight	Height
74	170 cm
80	180 cm
75	175.5 cm
-	

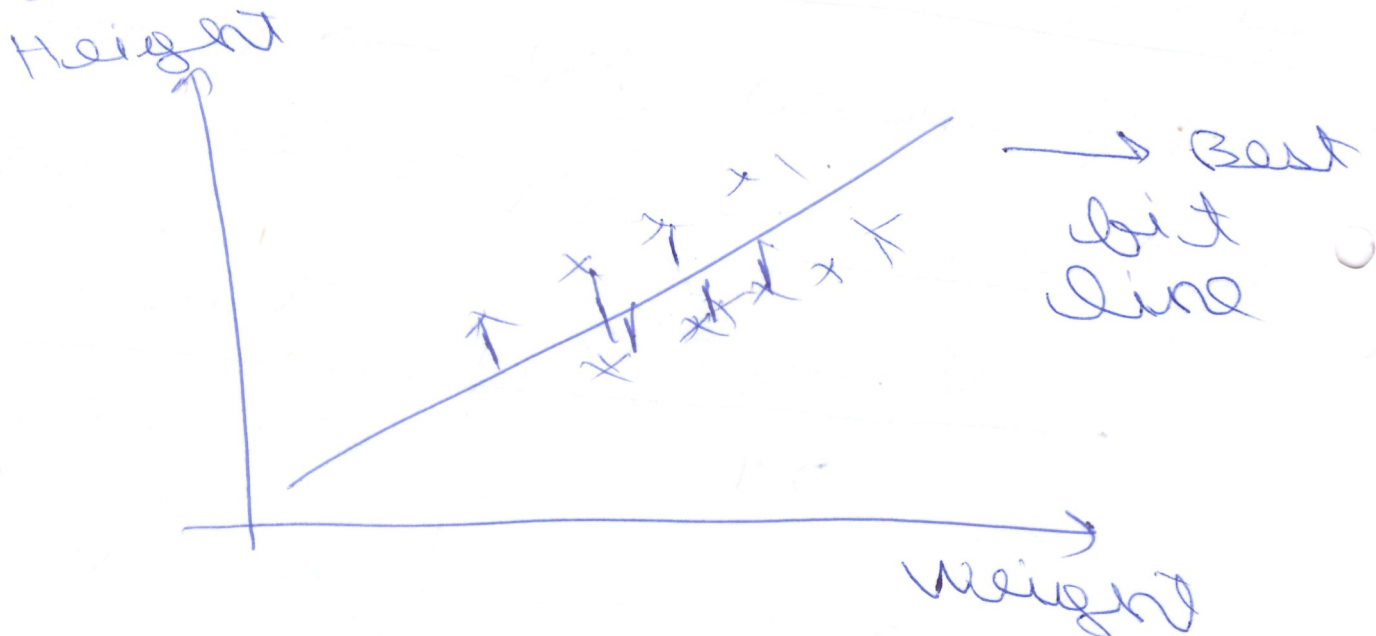


weight are Independent Features

Height is Dependent Feature

- If we have one input feature, it's simple linear regression

$\hookrightarrow$ If there are multiple input features, then it's multiple linear regression

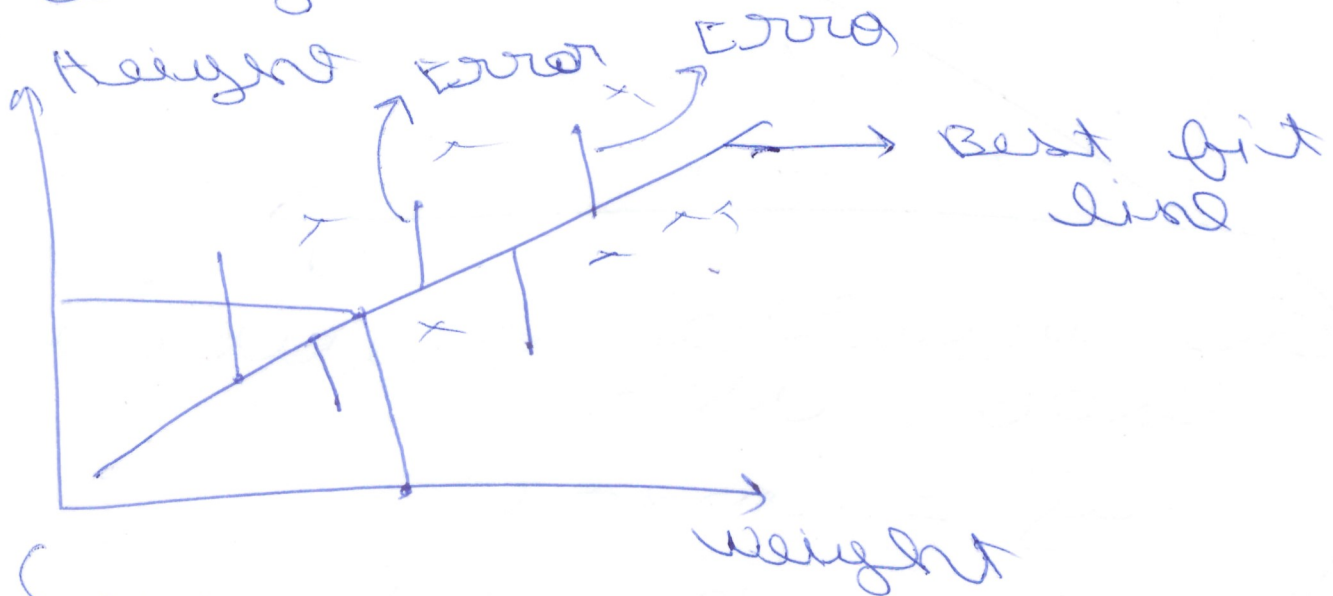


The best fit line helps in further prediction

- The distance of predicted & actual must be minimal

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So when we get a new
weight



→ We see where that point
is meeting the y-axis &
we get the height.



The equation represented here is:-

$$y = mx + c$$

$$\text{Intercept} = \beta_0 + \beta_1 x$$

~~$$h_0(x) = 0$$~~

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$$\theta_0 = \text{Intercept}$$

$$\text{if } x = 0$$

$$h_0(x) = \theta_0$$

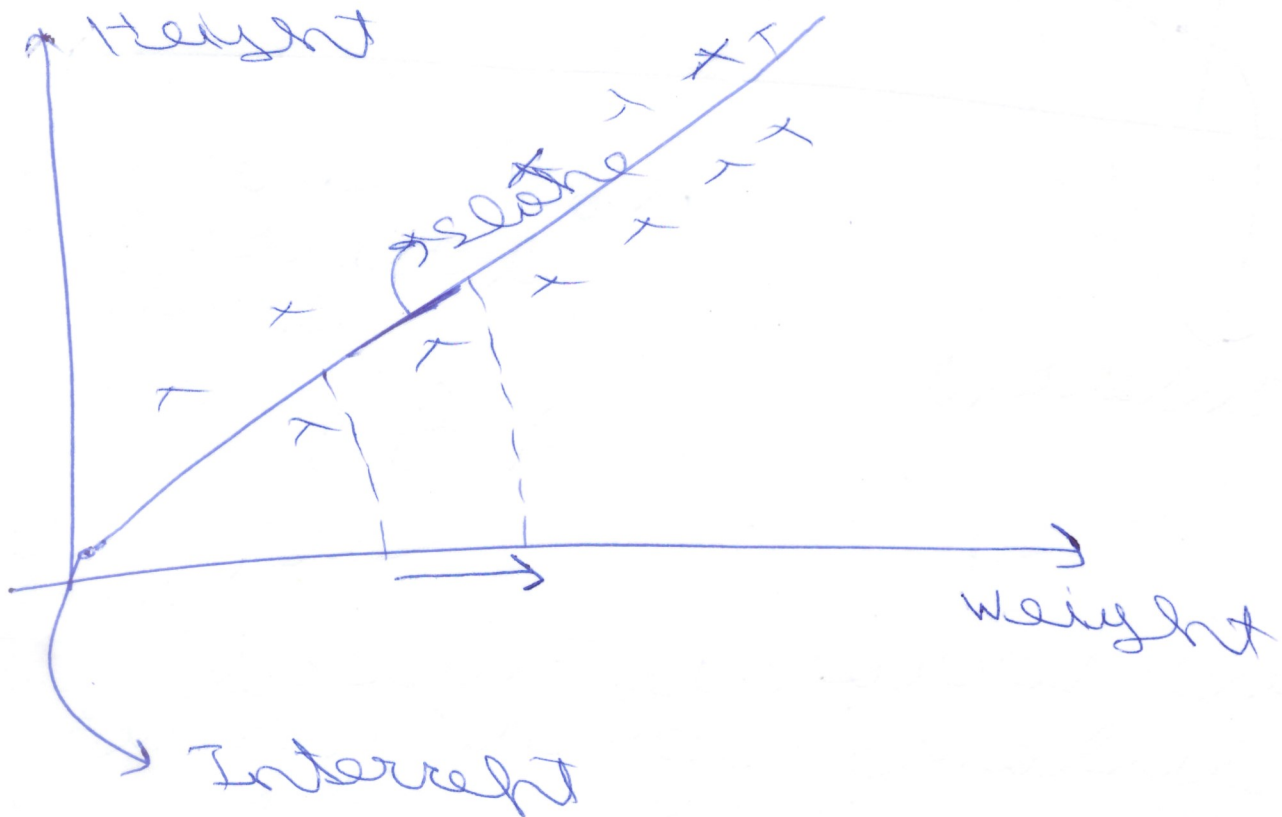
$$\theta_0 = \text{Intercept}$$

$$\theta_1 = \text{slope or coefficient}$$

~~sub has the~~

(5) Since x is temperature then y is weight, so

→ And we try plotting y vs x in y axis.

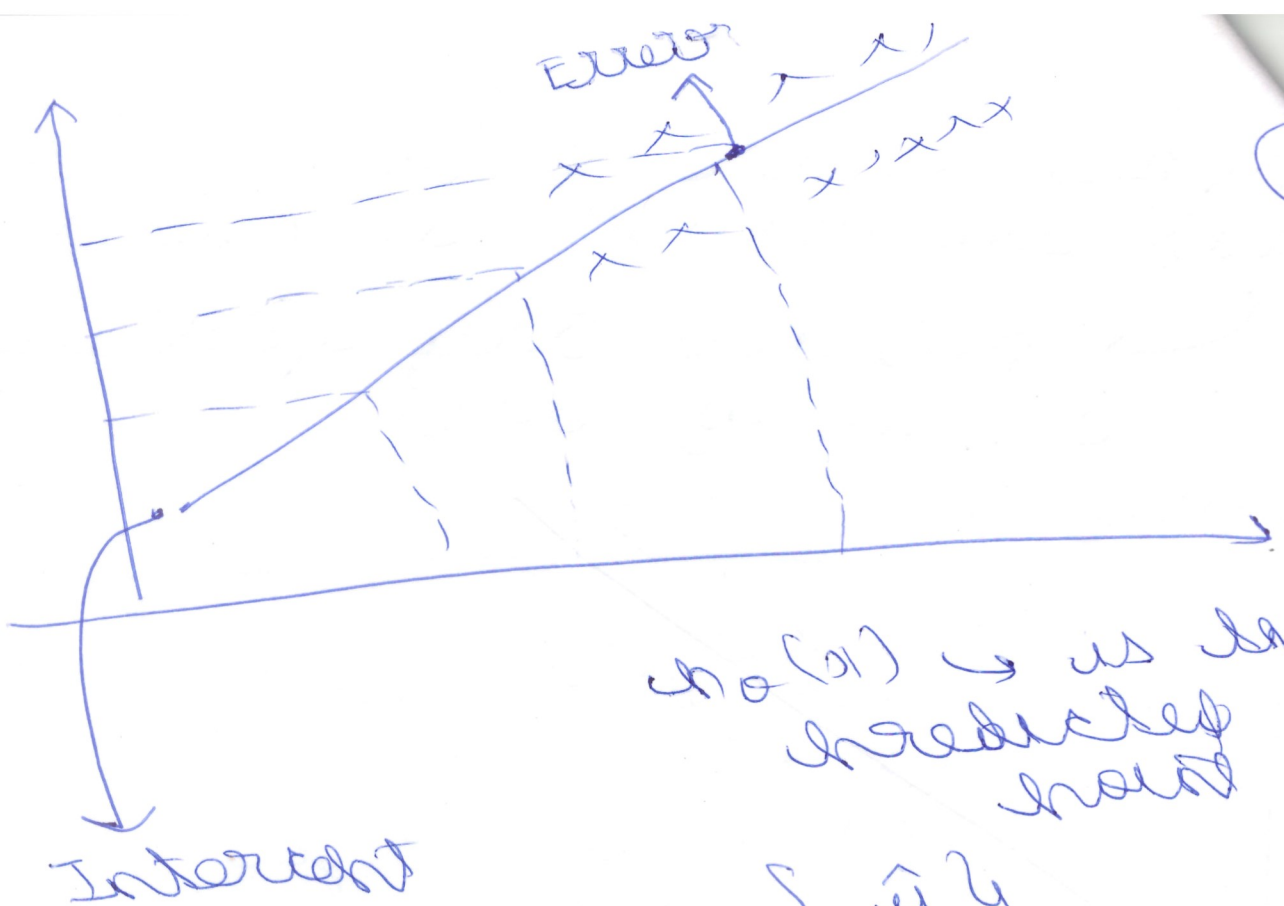


~~if~~ slope is represented by θ_1

~~$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$~~

if $x = 0$

$$h_0(x) = \theta_0$$



$x_0(x)$ → as the
predicted
point

$\{ \hat{y} \}$

→ predicted
point

Zero Error is basically the
distance or difference
between some point & predicted
point

$x_0(x)$ → Predicted
Point
denoted as
 $\{ \hat{y} \}$

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Cost Function :-

The cost function is represented as :-

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (\underbrace{\text{Predicted}}_{\hat{y}^{(i)}} - \underbrace{\text{Actual}}_{y^{(i)}})^2$$

Mean Squared Error

As subtraction it gives error
As the summation of all error, the ~~error~~ ~~sum~~ errors must be minimized
here, we apply cost function

④ The Final Aim required

We need to solve :-

Our main aim is to minimise the cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x_i) - y_i)^2$$

we get the mean squared error.

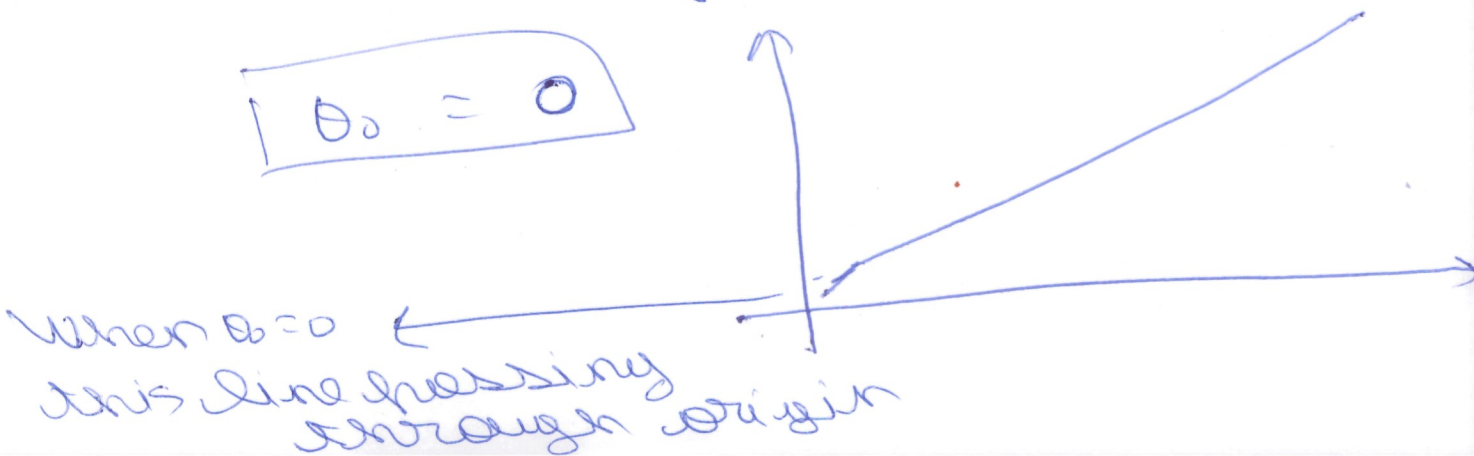
We continuously need to get the values of θ_0, θ_1 , where we find the best fit line for prediction

→ The summation should be minimal.

① $h_0(x) = \theta_0 + \theta_1 x$

→ Since there are 2 variables

$\theta_0 = 0$



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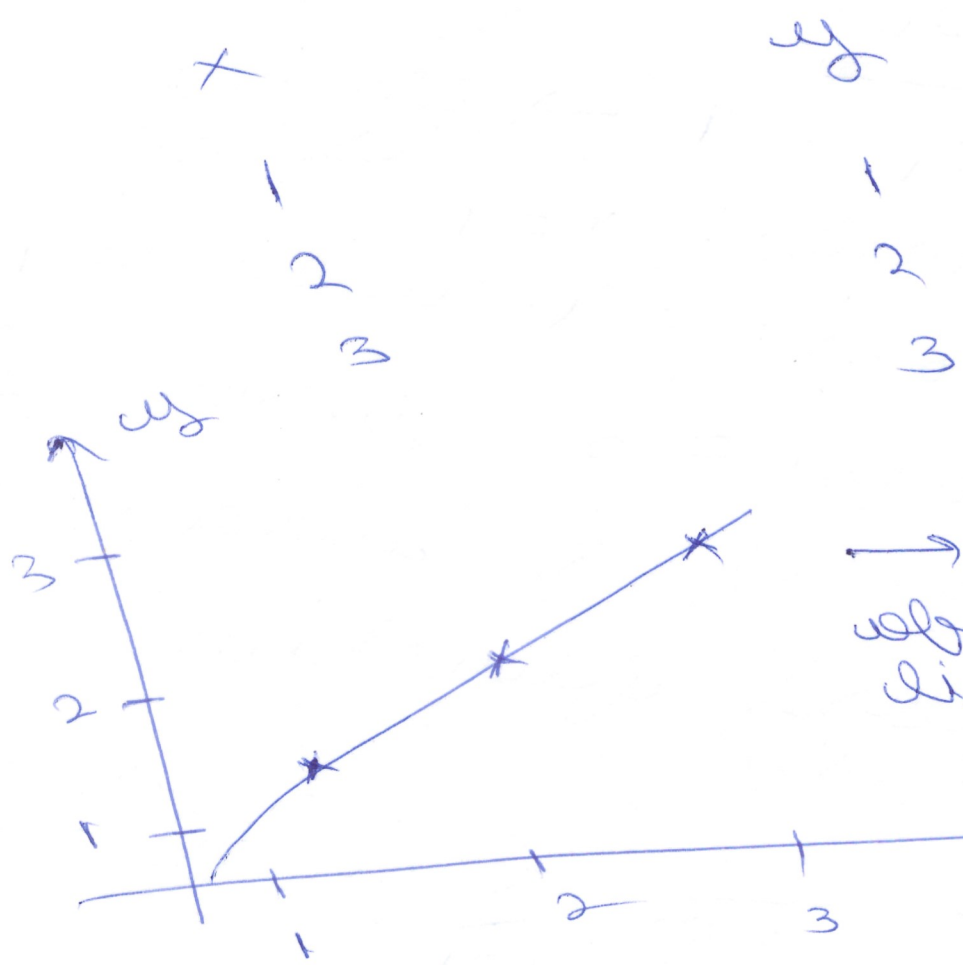
the intercept is exactly zero.
 $\theta_0 = 0$

The equation becomes:

$$h_0(x) = \theta_1 x$$

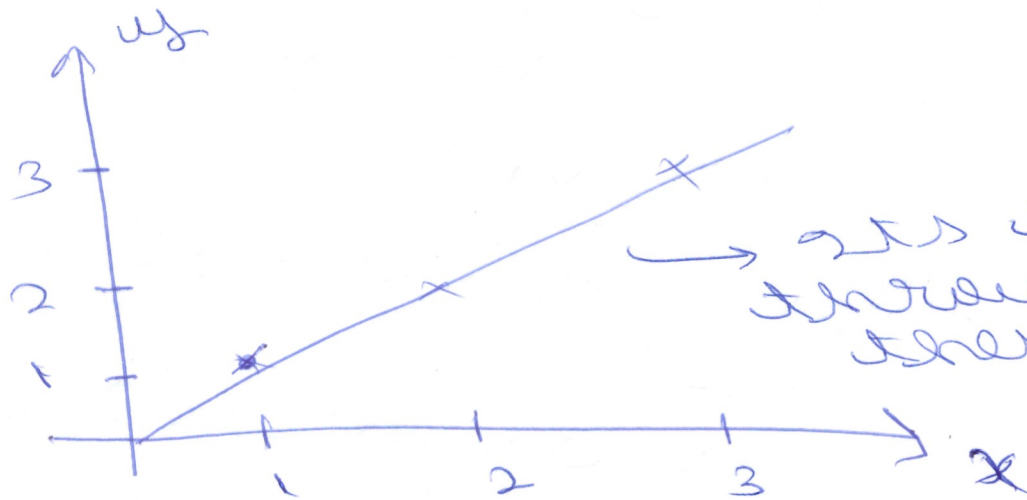
Now comparing the values of x & y

The dataset is presented as :-



→ Equation of straight line used

$$h_{\theta}(x) = \theta_1 x$$



$$h_{\theta}(x) = \theta_1 x$$

let $\theta_1 = 1 \rightarrow$ this is the slope

then when $x=1$

$$\hookrightarrow h_{\theta}(1) = 1 \quad \text{when } x=1$$

$$h_{\theta}(x) = 2 \quad \text{when } x=2$$

$$h_{\theta}(x) = 3 \quad \text{when } x=3$$

the cost function will be

$$J(\theta_1) = \frac{1}{2 \times 3} \sum_{i=1}^m (h_{\theta}(x)^{(i)} - y^{(i)})^2$$

$$= \frac{1}{2 \times 3} [(1-1)^2 + (2-2)^2 + (3-3)^2]$$

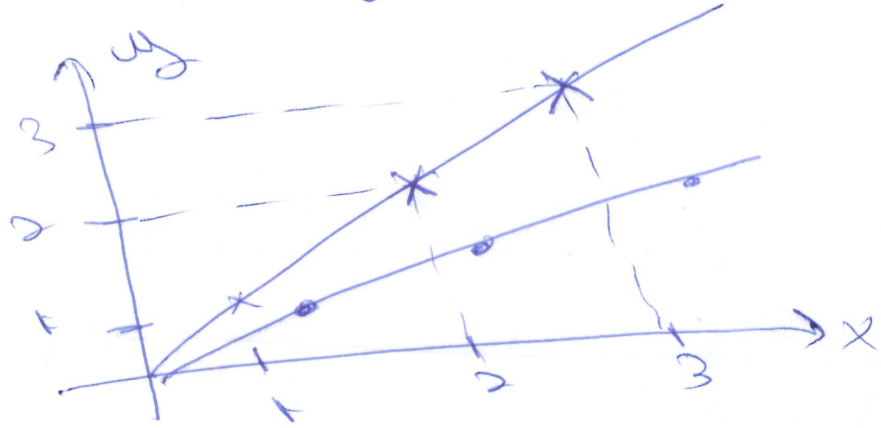
$$\hookrightarrow J(\theta_1) = 0$$

Let $\theta_1 = 0.5$

$\log(x) = 0.5$ at $x=1$

$\log(x) = 1$ at $x=2$

$\log(x) = 0.1x$



For $x = 0.5$, we do it as

$$J(\theta_1) = \frac{1}{2 \times 3} \left[(0.5 - 1)^2 + (1 - 2)^2 + (1.5 - 3)^2 \right]$$

$\theta_1 = 0.5$

$J(\theta_1) \approx 0.58$

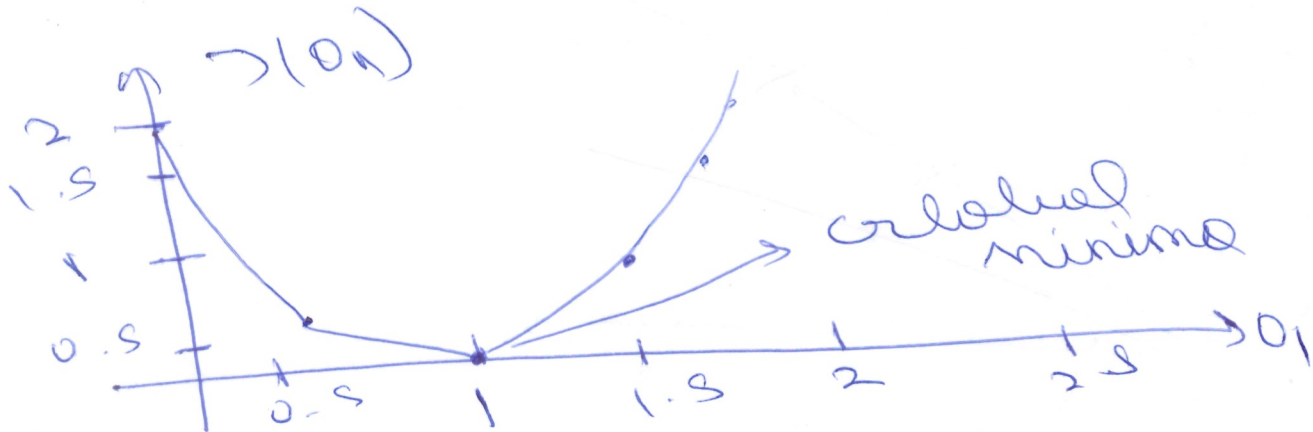
2b $\theta_1 = 0$

(12)

$$J(\theta_1) = \frac{1}{2 \times 3} [(0-1)^2 + (0-2)^2 + (0-3)^2]$$

$$J(\theta_1) \approx 2.3$$

$$J(\theta_1) \approx 2.3$$



→ The error has definitely minimized

→ Because we're getting our best fit line

⇒ Hence this is called as Gradient Descent. is also used in Deep Learning techniques

• We come to this specific point by changing θ_1 & θ_0 values.